

Summary of Topic 1 terms

program

A **program** consists of instructions executing one at a time. A Java program starts in the `main()` method, and executes the statements within `main`'s braces `{ }`, one at a time.

process

When a program performs computations on data, such as adding two values like `x + y`.

input

Information that a program gets from the keyboard, or a file, touchscreen, network, etc.

output

Information that a program sends somewhere, such as to a display screen, or to a file, network, etc.

variable

Used to store data within a program.

semicolon

Used to end a Java program statement (like a period for an English sentence).
Each Java statement typically appears alone on a line and ends with a semicolon.

Scanner

A text parser that can read numbers, words, or phrases from an input source, such as the keyboard.

nextInt()

Method used with a `Scanner` object to read an integer from input.
For example, given `Scanner` object `scnr`, the following statement gets an integer input value and assigns variable **num** with that value: `num = scnr.nextInt();`

System.out.print() / System.out.println()

Used to send output from a Java program to the display screen.

string literal

Text in double quotes (`" "`). For example: `"Hello"`

newline

Produces a new output line
Can be used by inserting `\n`, known as a newline character, within a string literal.

comment

A comment is text a programmer adds to code, to be read by humans to better understand the code but ignored by the compiler.

single-line comment

A single-line comment starts with `//` and includes all the following text on that one line.

multi-line comment (block comment)

A multi-line comment starts with `/*` and ends with `*/`, where all text lines between `/*` and `*/` is part of the comment.

Whitespace

Whitespace refers to blank space in program code, input, or output (space, tab, and newline characters).

hardware

Physical parts of a computer.

software

programs (the 0's and 1's) that instruct the computer what to do

transistors

Small switches that are integrated onto a silicon chip.

integrated circuit

A silicon chip with millions of tiny electrical components

central processing unit (CPU) / processor

The integrated circuit that implements the “brain” of the computer. The CPU runs the computer's programs, reading and executing instructions from memory, performing operations, and reading/writing data from/to memory.

graphics processing unit (GPU)

Additional processor that interfaces with the CPU to handle graphics processing for a computer.

arithmetic logic unit (ALU)

performs arithmetic (+, -, *, /) and logic (AND, OR, XOR, NOT) operations on binary integers.

control unit

tells computer components, such as memory, the ALU, and input and output devices, how to respond to computer instructions.

volatile

Storage device that loses all stored data when electricity is turned off

non-volatile

Storage device that maintains stored data even when electricity is turned off

memory

hardware that can store 0s and 1s in each of a series of thousands of addressed locations

RAM (random-access memory) / main memory

volatile memory that temporarily holds data and instructions read from disk, to be used by the CPU when running programs

cache (pronounced: cash)

an additional faster, but smaller, type of memory that stores copies of most frequently used instructions and data to provide quicker access to them

secondary storage devices

long term, non-volatile file storage

hard disk drive (HDD)

secondary storage drive that stores files and other data using a spinning disk to store individual bits via magnetism

solid-state drive (SSD)

secondary storage drive that uses a computer chip (with no moving parts) instead of a spinning disk to store files and other data

flash

A particular kind of non-volatile storage technology with the feature that large blocks of data can be erased simultaneously (in a "flash").

operating system

the software that supports a computer's basic functions, such as scheduling tasks, executing applications, and controlling peripherals

Keyboard

Hardware that allows a user to type characters. Allows a user to provide input to the computer.

Screen

Hardware for output that displays characters and images to a user.

clock

ticks at a specific frequency and controls the execution rate (speed) of a processor.

gigahertz (GHz)

One gigahertz is 1 billion clock ticks per second.

machine language

Programming language that stores instructions using 0s and 1s

High-level language

Programming language that is more "English-like", used by humans

compilers

Programs that translate high-level language programs into machine language that can be executed on a computer

operating system

the software that supports a computer's basic functions, such as scheduling tasks, executing applications, and controlling peripherals. Provides the interface between the user and the computer.

Java

In 1991, James Gosling and a team of engineers at Sun Microsystems (acquired by Oracle in 2010) began developing the Java language with the intent of creating a language whose executable could be ported to different processors, with the first public release in 1995.

problem solving

Creating a methodical solution to a given task.

decomposition

Breaking down a complex problem into a series of smaller or simpler problems.

pattern recognition

Identification of common characteristics among different problems and within the same problem.

algorithms

Sequences of instructions used to solve problems or perform tasks in a structured manner.

abstraction

Abstraction is the generalization of a problem so that a solution can be applied to different specific problems.

computational thinking

The thought processes by which a problem is formulated and solved with correct, precise, logical programs

bit

0 or 1 (binary digits)

byte

A group of 8 bits

word

An even multiple of bytes (containing some power of 2 bits – 8, 16, 32, 64, etc)

Most current computers support a word size of either 32 or 64 bytes, and the machine language and hardware for the computer are designed around that word size.

Kilobyte (KB)

approximately one thousand bytes ($2^{10} = 1024$ bytes)

Megabyte (MB)

approximately one million bytes (2^{20} bytes)

Gigabyte (GB)

approximately one billion bytes (2^{30} bytes)

Terabyte (TB)

approximately one trillion bytes (2^{40} bytes)

register

Small storage locations within the CPU, which can generally store between 32 and 128 word-sized values when the CPU is running programs.